

UC SAN DIEGO / ECE 251C

3

Finite-energy signals and sampling

Signal energy, Fourier identities, sampling, and aliasing

When do equally spaced samples determine a signal uniquely?

Sampling connects continuous- and discrete-domain signal processing: from a function $f(t)$, we form the sequence $f[n] := f(nT)$, where $T > 0$ is the sampling step. We first introduce signal energy and inner products, then use Parseval and Plancherel to connect them to Fourier analysis. The Dirac comb then reveals spectral repetition, aliasing, and ideal low-pass reconstruction.

3.1 Finite-energy sequences: the space $\ell^2(\mathbb{Z})$

For a finite vector $x \in \mathbb{C}^N$, its squared Euclidean norm is $\sum_{n=0}^{N-1} |x[n]|^2$. The same sum defines the *energy* of a sequence indexed by all integers:

$$\|x\|_{\ell^2}^2 := \sum_{n \in \mathbb{Z}} |x[n]|^2, \quad \ell^2(\mathbb{Z}) := \left\{ x : \mathbb{Z} \rightarrow \mathbb{C} : \sum_{n \in \mathbb{Z}} |x[n]|^2 < \infty \right\}. \quad (3.1)$$

The norm $\|x\|_{\ell^2}$ is the square root of the energy. The subscript names the space; it distinguishes an infinite sequence from a finite vector or a function on $\mathbb{R}$. A finite vector becomes an ℓ^2 sequence by setting all other entries to zero.

An infinite sequence can have finite energy. For $x[n] = 2^{-|n|}$,

$$\|x\|_{\ell^2}^2 = 1 + 2 \sum_{n=1}^{\infty} 4^{-n} = 1 + 2 \frac{1/4}{1 - 1/4} = \frac{5}{3}.$$

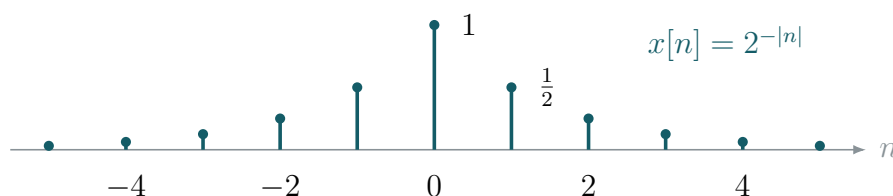


Figure 3.1: Infinitely many nonzero entries, but a finite sum of squared magnitudes. Finite energy is not the same as finite support.

3.2 Inner products of sequences

For real finite vectors, $\langle x, y \rangle = x^\top y$. For $x, y \in \ell^2(\mathbb{Z})$, use

$$\langle x, y \rangle_{\ell^2} := \sum_{n \in \mathbb{Z}} x[n]y^*[n], \quad \|x\|_{\ell^2} = \sqrt{\langle x, x \rangle_{\ell^2}}. \quad (3.2)$$

The superscript $*$ denotes complex conjugation: $(a + jb)^* = a - jb$ for real a, b . In this convention the inner product is *linear in its first argument* and conjugate-linear in its second:

$$\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle, \quad \langle x, ay \rangle = a^*\langle x, y \rangle.$$

Since $x[n]x^*[n] = |x[n]|^2$, we have $\langle x, x \rangle_{\ell^2} = \|x\|_{\ell^2}^2$. In other words, the inner product *induces the ℓ^2 norm*. Without conjugation, a sequence with only $x[0] = j$ nonzero would give -1 instead of energy one.

3.3 Finite-energy functions: the space $L^2(\mathbb{R})$

For a continuous-domain signal, replace the sum by an integral:

$$\|f\|_{L^2}^2 := \int_{\mathbb{R}} |f(t)|^2 dt, \quad L^2(\mathbb{R}) := \left\{ f : \mathbb{R} \rightarrow \mathbb{C} : \int_{\mathbb{R}} |f(t)|^2 dt < \infty \right\}. \quad (3.3)$$

We assume the integrals are defined. The inner product induces the norm:

$$\langle f, g \rangle_{L^2} := \int_{\mathbb{R}} f(t)g^*(t) dt, \quad \|f\|_{L^2} = \sqrt{\langle f, f \rangle_{L^2}}. \quad (3.4)$$

Indeed, setting $g = f$ gives $\langle f, f \rangle_{L^2} = \int_{\mathbb{R}} |f(t)|^2 dt = \|f\|_{L^2}^2$, just as for sequences. The argument t can denote time or space; a *continuous-domain* signal need not be a continuous function.

Example: a pulse with infinite extent. For $f(t) = e^{-|t|}$,

$$\|f\|_{L^2}^2 = 2 \int_0^\infty e^{-2t} dt = 2 \left[-\frac{1}{2}e^{-2t} \right]_0^\infty = 1.$$

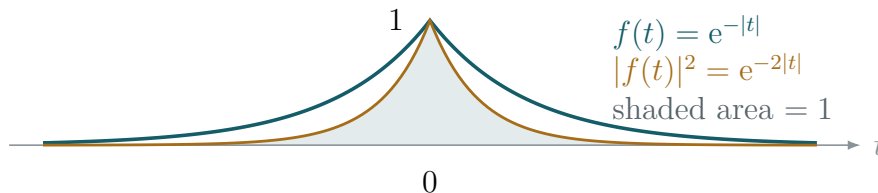


Figure 3.2: Energy is area under the *squared magnitude*, not area under the signal itself.

A nonzero constant or a persistent sinusoid has infinite energy on $\mathbb{R}$. These signals remain useful for studying frequency response, but are not in $L^2(\mathbb{R})$.

3.4 Other measures of size: ℓ^p and L^p

For $1 \leq p < \infty$, the corresponding definitions are

$$\|x\|_{\ell^p} := \left(\sum_{n \in \mathbb{Z}} |x[n]|^p \right)^{1/p}, \quad \|f\|_{L^p} := \left(\int_{\mathbb{R}} |f(t)|^p dt \right)^{1/p}. \quad (3.5)$$

The spaces $\ell^p(\mathbb{Z})$ and $L^p(\mathbb{R})$ consist of signals with finite respective norms. The case $p = 1$ measures absolute summability or integrability; the case $p = 2$ gives the square root of energy. We will return to absolute-value costs when studying sparsity.

At $p = \infty$, use the smallest uniform bound:

$$\|x\|_{\ell^\infty} = \sup_{n \in \mathbb{Z}} |x[n]|, \quad \|f\|_{L^\infty} = \operatorname{ess\,sup}_{t \in \mathbb{R}} |f(t)|.$$

A *supremum* is the least upper bound. It need not be an attained maximum: the sequence $x[n] = 1 - 1/(n + 1)$ for $n \geq 0$, extended by zero for $n < 0$, has supremum one but no entry equal to one.

Functions that differ only on a set of total length zero agree *almost everywhere* and represent the same element of L^p . An *essential supremum* is the smallest bound allowed to fail on such a set. For example, let $g(t) = 1$ on $[-1, 1]$, set $g(0) = 3$, and set $g(t) = 0$ outside the interval. Then

$$\sup_t |g(t)| = 3, \quad \|g\|_{L^\infty} = 1, \quad \|g\|_{L^2}^2 = 2.$$

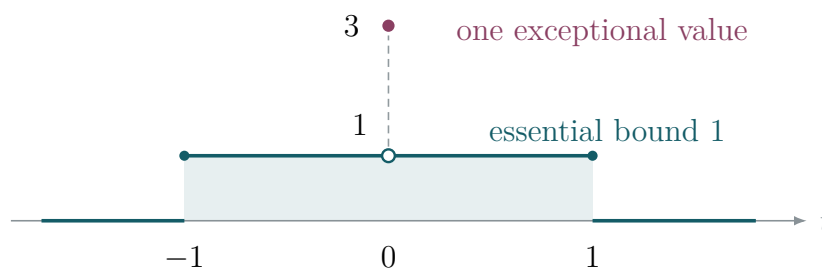


Figure 3.3: An isolated value changes the ordinary supremum, but not an integral or the essential supremum. The dashed guide is not part of the signal.

Why this matters for sampling

Energy alone does not specify point values. In ideal sampling we take f to be continuous, so the sample values $f[n] = f(nT)$ are unambiguous.

3.5 Parseval and Plancherel for sequences

Fourier analysis preserves inner products and energy, with constants determined by the transform convention. For sequences we write

$$\hat{x}(\theta) = X(e^{j\theta}) = \sum_{n \in \mathbb{Z}} x[n] e^{-jn\theta}, \quad -\pi \leq \theta \leq \pi. \quad (3.6)$$

We use θ for the *discrete-domain* angular frequency. The spectrum repeats every 2π because $e^{-jn(\theta+2\pi)} = e^{-jn\theta}$ for integer n .

The calculation for finite sequences. For integers m, n ,

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j(n-m)\theta} d\theta = \begin{cases} 1, & n = m, \\ 0, & n \neq m. \end{cases}$$

For $n = m$ we integrate one. Otherwise the antiderivative is $e^{-j(n-m)\theta} / [-j(n-m)]$, with equal values at both endpoints. Therefore

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\theta) \hat{y}^*(\theta) d\theta &= \sum_n \sum_m x[n] y^*[m] \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-j(n-m)\theta} d\theta \\ &= \sum_n x[n] y^*[n] = \langle x, y \rangle_{\ell^2}. \end{aligned} \quad (3.7)$$

All sums are finite here. Setting $y = x$ proves the energy identity. The same integral identity extracts one coefficient:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \hat{x}(\theta) e^{jn\theta} d\theta. \quad (3.8)$$

The finite-energy identities. Parseval and Plancherel extend the displayed identities to all $x, y \in \ell^2(\mathbb{Z})$. In particular,

$$\boxed{\|x\|_{\ell^2}^2 = \frac{1}{2\pi} \|\hat{x}\|_{L^2[-\pi, \pi]}^2} \quad (3.9)$$

The Fourier transform identifies finite-energy sequences with spectra in $L^2[-\pi, \pi]$. For a sequence, we measure spectral energy over *one period*, not over the whole real line.

3.6 The continuous-domain energy identity

Our angular-frequency convention is

$$\hat{f}(\omega) = \int_{\mathbb{R}} f(t) e^{-j\omega t} dt, \quad f(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega) e^{j\omega t} d\omega. \quad (3.10)$$

Theorem 3.1 (Parseval and Plancherel). The Fourier transform extends to finite-energy signals. For $f, g \in L^2(\mathbb{R})$,

$$\langle f, g \rangle_{L^2(\mathbb{R})} = \frac{1}{2\pi} \langle \hat{f}, \hat{g} \rangle_{L^2(\mathbb{R})}, \quad (3.11)$$

$$\|f\|_{L^2}^2 = \frac{1}{2\pi} \|\hat{f}\|_{L^2}^2. \quad (3.12)$$

Every L^2 spectrum is the transform of a unique L^2 signal, with equality of functions understood almost everywhere.

Where the identity comes from. For $f, \hat{g} \in L^1(\mathbb{R})$ with Fourier inversion valid for g , substitute the inverse transform into the inner product:

$$\begin{aligned} \int_{\mathbb{R}} f(t) g^*(t) dt &= \int_{\mathbb{R}} f(t) \left[\frac{1}{2\pi} \int_{\mathbb{R}} \hat{g}^*(\omega) e^{-j\omega t} d\omega \right] dt \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \left[\int_{\mathbb{R}} f(t) e^{-j\omega t} dt \right] \hat{g}^*(\omega) d\omega \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\omega) \hat{g}^*(\omega) d\omega. \end{aligned}$$

This gives the inner-product identity. Taking $g = f$ gives the energy identity. The theorem states that both formulas hold for all finite-energy signals.

Check the normalization. For $x[n] = \delta[n] + \delta[n-1]$, the DTFT is $1 + e^{-j\theta}$, so

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |1 + e^{-j\theta}|^2 d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} (2 + 2\cos\theta) d\theta = 2 = \|x\|_{\ell^2}^2.$$

For the unit box from Lecture 2, $\|h\|_{L^2}^2 = 1$ and $\hat{h}(\omega) = \text{sinc}(\omega/(2\pi))$, where $\text{sinc}(u) = \sin(\pi u)/(\pi u)$ with value one at zero. Plancherel and $u = \omega/(2\pi)$ give

$$1 = \frac{1}{2\pi} \int_{\mathbb{R}} \left| \text{sinc}\left(\frac{\omega}{2\pi}\right) \right|^2 d\omega = \int_{\mathbb{R}} |\text{sinc}(u)|^2 du.$$

3.7 Ideal sampling: multiplication by a Dirac comb

Let f be continuous and let $T > 0$ be the sampling step. We sample at the locations nT , $n \in \mathbb{Z}$. Define the *Dirac comb*

$$\mathbb{I}_T(t) := \sum_{n \in \mathbb{Z}} \delta(t - nT). \quad (3.13)$$

It is a train of unit-weight impulses, spaced T apart. On any bounded interval there are only finitely many impulses. Ideal sampling multiplies the input by this comb. By the sifting property,

$$\begin{aligned} f_T(t) &:= f(t)\mathbb{I}_T(t) = \sum_{n \in \mathbb{Z}} f(t)\delta(t - nT) \\ &= \sum_{n \in \mathbb{Z}} f[n]\delta(t - nT), \quad f[n] = f(nT). \end{aligned} \quad (3.14)$$

The value $f(nT)$ is the *weight* of the impulse at nT .

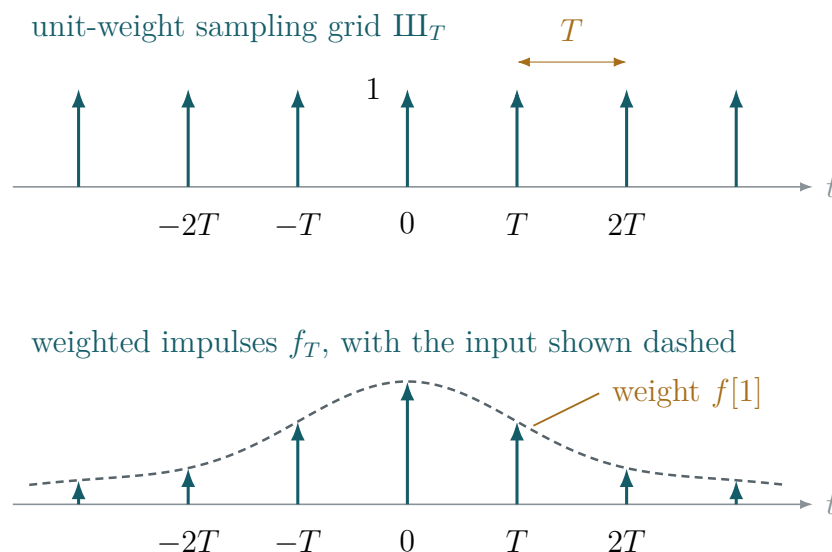


Figure 3.4: Multiplication by the comb retains the values on the grid. The input between samples is not recorded. The impulse train is not a staircase or a line joining the samples.

The *sample sequence* $f[n]$ lives on $\mathbb{Z}$, while the impulse train f_T lives on $\mathbb{R}$. They encode the same numbers in different mathematical objects. A Dirac comb or a nonzero sampled impulse train is not an ordinary L^2 function: we do not square impulses to assign it an energy.

3.8 The spectrum of the sampled signal is periodic

Applying the Fourier transform to [equation \(3.14\)](#) gives

$$\begin{aligned}\hat{f}_T(\omega) &= \int_{\mathbb{R}} \sum_{n \in \mathbb{Z}} f[n] \delta(t - nT) e^{-j\omega t} dt \\ &= \sum_{n \in \mathbb{Z}} f[n] \int_{\mathbb{R}} \delta(t - nT) e^{-j\omega t} dt \\ &= \sum_{n \in \mathbb{Z}} f[n] e^{-j\omega nT}.\end{aligned}\tag{3.15}$$

The sampling angular frequency is $\omega_0 := 2\pi/T$. Since n is an integer,

$$e^{-j(\omega+\omega_0)nT} = e^{-j\omega nT} e^{-j2\pi n} = e^{-j\omega nT}.$$

Consequently, $\hat{f}_T(\omega + \omega_0) = \hat{f}_T(\omega)$: the spectrum repeats every $2\pi/T$.

Physical and normalized frequency. The sampled sequence has DTFT

$$F(e^{j\theta}) = \sum_{n \in \mathbb{Z}} f[n] e^{-jn\theta}.$$

This is distinct from the continuous-domain transform $\hat{f}(\omega)$. Comparing it with [equation \(3.15\)](#) gives

$$\boxed{\hat{f}_T(\omega) = F(e^{j\omega T}), \quad \theta = \omega T.}\tag{3.16}$$

The variable ω measures radians per unit of the continuous coordinate; θ measures radians per sample. They coincide only when $T = 1$.

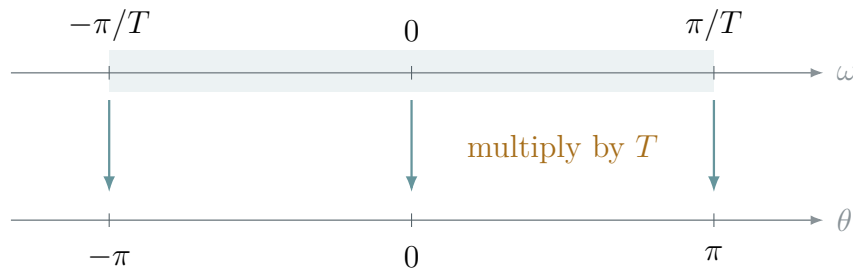


Figure 3.5: The base interval $[-\pi/T, \pi/T]$ has width $2\pi/T$. In the normalized variable $\theta = \omega T$, it becomes $[-\pi, \pi]$.

A nonzero periodic spectrum has infinite energy when integrated over *all* of $\mathbb{R}$. For a sequence we integrate over just one period, as in [equation \(3.9\)](#). There is no conflict with the continuous-domain theorem: the sampled object f_T is an impulse train, not an $L^2(\mathbb{R})$ function.

Periodicity tells us *where* the spectrum repeats. To determine *what* is repeated, we next compute the Fourier transform of the sampling comb.

3.9 The Fourier transform of a Dirac comb

Recall the Fourier series of a T -periodic signal g :

$$g(t) = \sum_{k \in \mathbb{Z}} c_k e^{jk\omega_0 t}, \quad c_k = \frac{1}{T} \int_{-T/2}^{T/2} g(t) e^{-jk\omega_0 t} dt.$$

The comb has one impulse, at zero, in this central period. Its coefficients are therefore

$$c_k = \frac{1}{T} \int_{-T/2}^{T/2} \text{III}_T(t) e^{-jk\omega_0 t} dt = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T}.$$

Hence $\text{III}_T(t) = T^{-1} \sum_k e^{jk\omega_0 t}$. This is an *impulse identity*: as with the Dirac impulse, it is understood through integration.

Transforming one exponential. The inverse transform of $2\pi\delta(\omega - a)$ is

$$\frac{1}{2\pi} \int_{\mathbb{R}} 2\pi\delta(\omega - a) e^{j\omega t} d\omega = e^{jat}.$$

Thus $e^{jat} \xleftrightarrow{\mathcal{F}} 2\pi\delta(\omega - a)$. This exchange is related to *Fourier duality*: whenever the ordinary integrals and inversion hold, inversion at $-\omega$ gives

$$f(t) \xleftrightarrow{\mathcal{F}} \hat{f}(\omega) \implies \hat{f}(t) \xleftrightarrow{\mathcal{F}} 2\pi f(-\omega).$$

Now transform the comb's Fourier series term by term as an impulse identity:

$$\boxed{\widehat{\text{III}}_T(\omega) = \frac{2\pi}{T} \sum_{k \in \mathbb{Z}} \delta\left(\omega - \frac{2\pi k}{T}\right).} \quad (3.17)$$

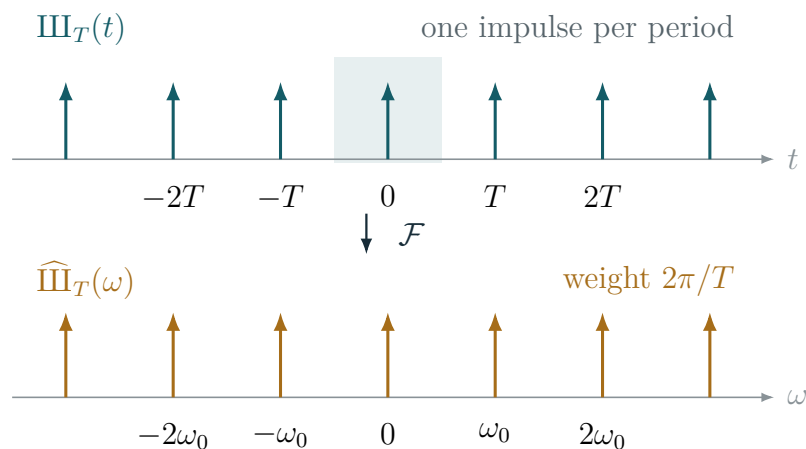


Figure 3.6: A comb transforms into a comb. Spatial spacing T becomes frequency spacing $2\pi/T$. The shaded interval on top is one sampling period; the arrows below have weight $2\pi/T$, not weight one.

3.10 Sampling periodizes the original spectrum

Multiplication in the signal domain becomes convolution in the Fourier domain, with our factor $1/(2\pi)$. For $f, \hat{g} \in L^1(\mathbb{R})$ with Fourier inversion valid for g , substitute the inverse transform of g to derive

$$\begin{aligned}\widehat{fg}(\omega) &= \int_{\mathbb{R}} f(t) \left[\frac{1}{2\pi} \int_{\mathbb{R}} \hat{g}(\nu) e^{j\nu t} d\nu \right] e^{-j\omega t} dt \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \hat{g}(\nu) \hat{f}(\omega - \nu) d\nu = \frac{1}{2\pi} (\hat{f} * \hat{g})(\omega).\end{aligned}$$

Apply the corresponding impulse identity with $g = \text{III}_T$. Convolution with a shifted impulse translates the spectrum, so [equation \(3.17\)](#) gives

$$\begin{aligned}\hat{f}_T(\omega) &= \frac{1}{2\pi} \left(\hat{f} * \frac{2\pi}{T} \sum_k \delta(\cdot - k\omega_0) \right) (\omega) \\ &= \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{f}(\omega - k\omega_0).\end{aligned}\tag{3.18}$$

Combining this with [equation \(3.15\)](#) yields the *Poisson summation formula* in the form needed here:

$$\boxed{\sum_{n \in \mathbb{Z}} f[n] e^{-j\omega n T} = \frac{1}{T} \sum_{k \in \mathbb{Z}} \hat{f}\left(\omega - \frac{2\pi k}{T}\right).}\tag{3.19}$$

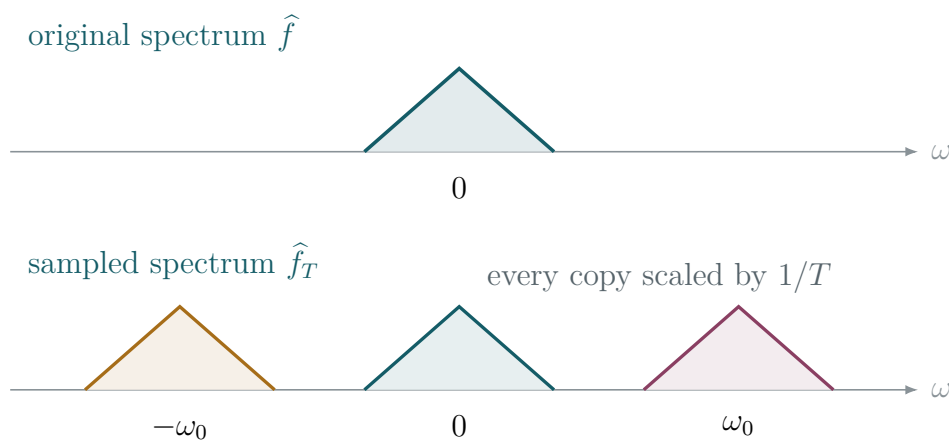


Figure 3.7: Sampling translates copies of the original spectrum by multiples of $\omega_0 = 2\pi/T$ and adds them. The vertical scales of the two rows differ by $1/T$. The triangle is an illustrative spectrum.

3.11 Aliasing is spectral overlap

Suppose $\hat{f}(\omega) = 0$ for $|\omega| > \omega_{\max}$. We call f *bandlimited* to $\omega_{\max}$. Its translated spectral copies have width $2\omega_{\max}$ and their centers are spaced $\omega_0 = 2\pi/T$ apart.

If $\omega_0 > 2\omega_{\max}$, neighboring copies are separated. If $\omega_0 < 2\omega_{\max}$, they can overlap. In an overlap region, the observed value is a *sum* of different spectral contributions. We generally cannot separate those contributions from their sum.

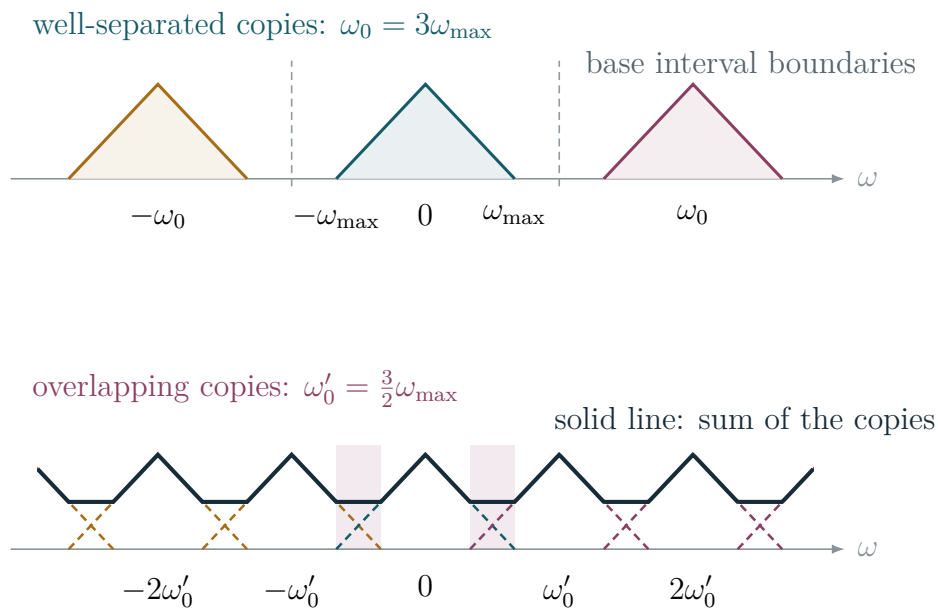


Figure 3.8: The same triangular spectrum sampled at two different steps. To compare the shapes, the rows show $T\hat{f}_T$ and $T'\hat{f}_{T'}$, respectively. Dashed curves in the lower row are individual copies; the solid curve is their sum. Shading marks overlap with the central copy.

Increasing T moves samples farther apart, but brings spectral copies *closer together*. This reversal is the key geometric fact. The familiar sufficient condition for an alias-free baseband is

$$\boxed{\omega_0 > 2\omega_{\max} \iff T < \frac{\pi}{\omega_{\max}}.} \quad (3.20)$$

If frequency is measured in cycles per unit, write $\nu_{\max} = \omega_{\max}/(2\pi)$ and $\nu_0 = 1/T$. The same inequality becomes $\nu_0 > 2\nu_{\max}$. The strict inequality leaves a gap between the copies.

3.12 Different functions can have exactly the same samples

Spectral overlap has a concrete consequence: the samples need not distinguish two different signals. Here is an example *within the finite-energy setting*. Let

$$q(t) = \text{sinc}\left(\frac{t}{4T}\right), \quad u(t) = q(t) \cos\left(\frac{2\pi t}{5T}\right), \quad v(t) = q(t) \cos\left(\frac{8\pi t}{5T}\right).$$

At every sampling location, the cosine factors agree:

$$\cos\left(\frac{8\pi n}{5}\right) = \cos\left(2\pi n - \frac{2\pi n}{5}\right) = \cos\left(\frac{2\pi n}{5}\right).$$

Consequently $u(nT) = v(nT)$ for every integer n . But at $t = T/2$, the two cosine factors are $\cos(\pi/5)$ and $\cos(4\pi/5) = -\cos(\pi/5)$; since $q(T/2) \neq 0$, the functions are different.

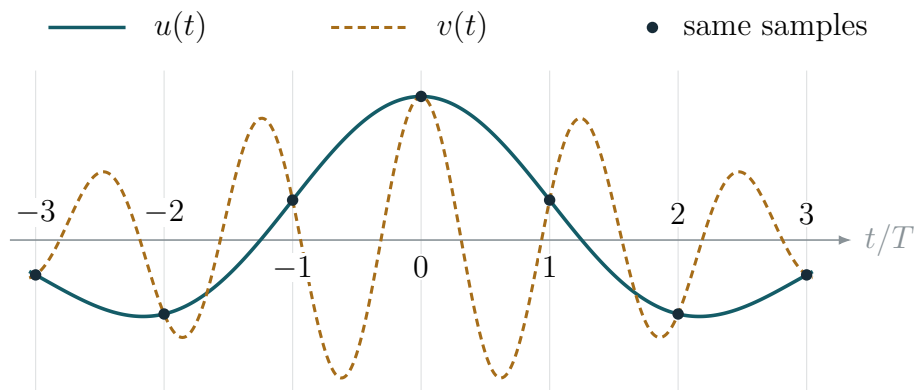


Figure 3.9: Two continuous functions pass through the same sample values but differ between samples. The solid and dashed curves are u and v ; the dots are their common samples.

These are not infinite-energy pure tones. Using $|\cos| \leq 1$ and the sinc energy computed earlier,

$$\|u\|_{L^2}^2, \|v\|_{L^2}^2 \leq \int_{\mathbb{R}} \left| \text{sinc}\left(\frac{t}{4T}\right) \right|^2 dt = 4T \int_{\mathbb{R}} |\text{sinc}(s)|^2 ds = 4T.$$

Sampling requires a signal model

A recovery theorem cannot hold for all finite-energy functions. A bandwidth restriction rules out ambiguous alternatives. The theorem will recover the function in the specified bandlimited class, not every function that happens to pass through the samples.

3.13 Recovering the spectrum by ideal low-pass filtering

Assume $\omega_{\max} < \pi/T$. On the base interval $[-\pi/T, \pi/T]$, the only possible nonzero term in equation (3.18) is the central copy. Thus

$$\hat{f}_T(\omega) = \frac{1}{T}\hat{f}(\omega) \quad (|\omega| \leq \pi/T),$$

apart from immaterial endpoint conventions. To extract this copy *and restore its scale*, use the response

$$H_T(\omega) := T\mathbf{1}_{[-\pi/T, \pi/T]}(\omega), \quad H_T(\omega)\hat{f}_T(\omega) = \hat{f}(\omega). \quad (3.21)$$

The gain is T , not one: sampling introduced a factor $1/T$. Outside the base interval the reconstruction filter removes every repeated copy.

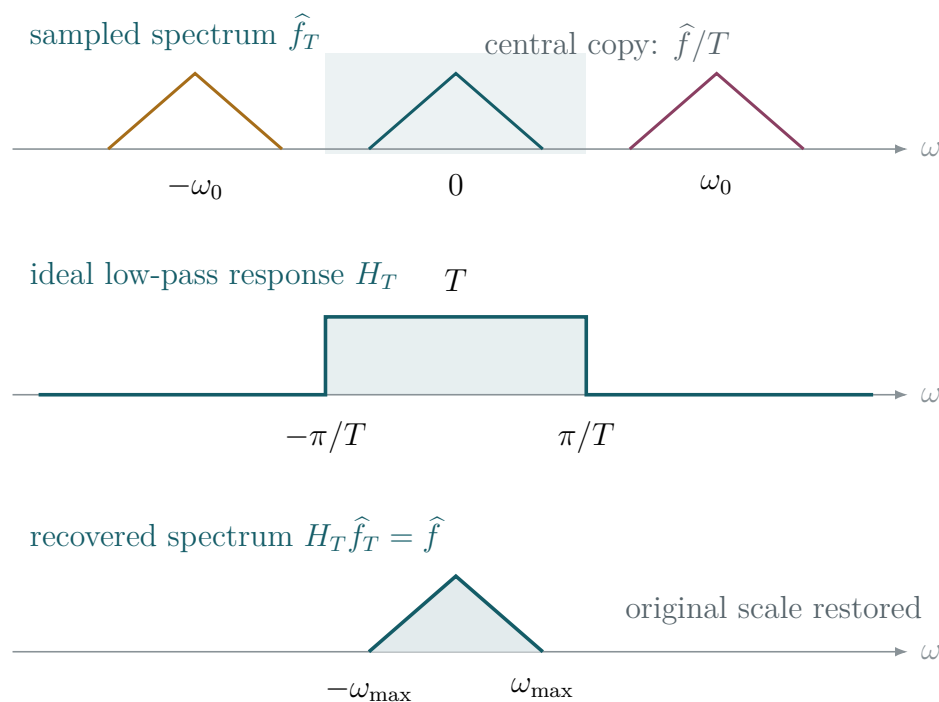


Figure 3.10: Keep one copy, remove the others, and undo the factor $1/T$. The shaded interval is the reconstruction passband. The rows use their own vertical scales.

The argument uses the bandwidth assumption twice: to prevent overlap before filtering, and to ensure the entire original spectrum lies inside the passband. A low-pass filter cannot unmix contributions that have already added within that band.

3.14 The reconstruction kernel is a sinc

Let h_T be the impulse response with Fourier transform H_T from equation (3.21). Invert the rectangle directly. For $t \neq 0$,

$$\begin{aligned} h_T(t) &= \frac{T}{2\pi} \int_{-\pi/T}^{\pi/T} e^{j\omega t} d\omega \\ &= \frac{T}{2\pi} \frac{e^{j\pi t/T} - e^{-j\pi t/T}}{jt} = \frac{T \sin(\pi t/T)}{\pi t} = \text{sinc}\left(\frac{t}{T}\right). \end{aligned} \quad (3.22)$$

At $t = 0$, the same integral is $(T/(2\pi))(2\pi/T) = 1$. Thus the formula includes zero when we define

$$\text{sinc}(s) := \begin{cases} \sin(\pi s)/(\pi s), & s \neq 0, \\ 1, & s = 0. \end{cases}$$

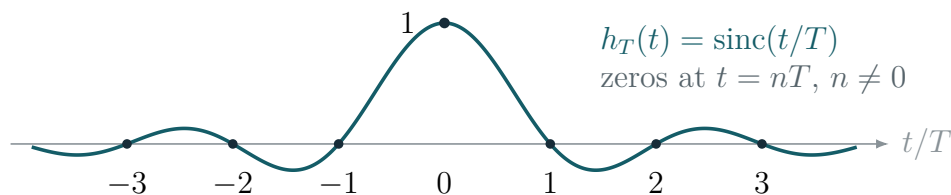


Figure 3.11: The reconstruction kernel is one at its own grid point and zero at every other grid point. The black dots mark its values on the sampling grid.

Filter the sampled impulse train. Each weighted impulse produces a shifted copy of h_T :

$$\begin{aligned} (h_T * f_T)(t) &= \sum_{n \in \mathbb{Z}} f[n] \int_{\mathbb{R}} h_T(t - u) \delta(u - nT) du \\ &= \sum_{n \in \mathbb{Z}} f[n] h_T(t - nT) \\ &= \sum_{n \in \mathbb{Z}} f[n] \text{sinc}\left(\frac{t - nT}{T}\right). \end{aligned} \quad (3.23)$$

At a grid point $t = mT$, every term except $n = m$ is zero. The remaining term equals $f[m] = f(mT)$. The kernel therefore *interpolates* the samples. The spectral argument supplies the stronger conclusion: for the stated bandlimited class, it recovers the function between the samples as well.